

Python Programming Assignment

Instructions:

- Solve all questions using Python.
- Use meaningful variable and function names.
- Write clean, well-commented code.
- Real-life logic and correct program flow are important.

1. Create a program that takes a user's name, age, and city and prints a formatted introduction message.
2. Write a program that calculates monthly electricity bills based on units consumed using conditional slabs.
3. Build a program that checks whether a given year is a leap year.
4. Create a grocery bill calculator where the user enters item prices until they type 'done'.
5. Write a program to count vowels and consonants in a sentence entered by the user.
6. Create a program that simulates a simple ATM menu (check balance, deposit, withdraw).
7. Write a program that finds the maximum and minimum number from a list entered by the user.
8. Create a student grading system using marks stored in a dictionary.
9. Write a program to remove duplicate elements from a list using sets.
10. Create a function that returns the factorial of a number using loops.
11. Write a program that checks if a password is strong based on given conditions.
12. Build a program that tracks daily expenses for a week and calculates total and average expense.
13. Write a program to generate Fibonacci numbers up to n terms using a function.
14. Create a program that simulates a simple voting system and counts votes using dictionaries.
15. Write a program that finds common elements between two lists.
16. Create a program to check if a number is prime using a function.
17. Write a program that converts temperatures between Celsius and Fahrenheit using functions.
18. Create a program that manages contact details (add, search, delete).
19. Write a program to analyze a list of salaries and calculate tax based on conditions.
20. Create a program that tracks attendance of students using sets.
21. Write a program that simulates a basic shopping cart system.
22. Create a function that checks whether a string is a palindrome.
23. Write a program that analyzes exam results and finds toppers.
24. Create a modular program for a library system (add books, issue books, return books).
25. Write a program that simulates a simple online food ordering system.